

Summary

An integrating sphere is a measurement tool used to measure the optical characteristics of a light source. The integrating sphere consists of a diffusely reflective surface with high reflectivity on the inner surface of the sphere. By repeatedly reflecting the light emitted from a light source inside the sphere as well as the light entering the sphere, the illumination on the inner wall is uniform, and by measuring a portion of this light, it serves as a tool for measuring the optical characteristics of the light source.

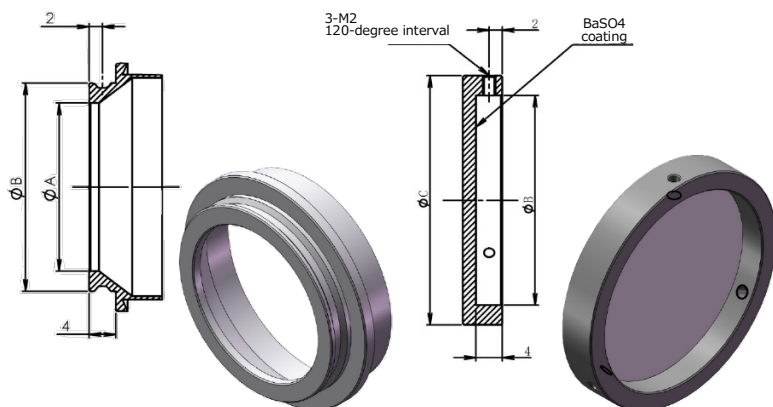
Application

- Measure characteristics such as color and light of the light source in a total luminous flux system
- Measure the transmittance and reflectance of diffusive objects
- Calibrate cameras and other devices by assembling them with a uniform light source

Model number	Inner diameter (inches)	Port Size (inches)			
		PO LE	0 degree	90 degrees	180 degrees
NIS-002-03A	2	0.5	-	0.5	0.5
NIS-003-03A	3	1	-	1	1
NIS-003-04A	3	1	1	1	1
NIS-004-03A	4	1	-	1	1.5
NIS-004-04A	4	1	1	1	1.5
NIS-006-03A	6	1	-	1	2
NIS-006-04A	6	1	1	1	2
NIS-008-03A	8	1	-	1	2.5
NIS-008-04A	8	1	1	1	2.5

Port and lid

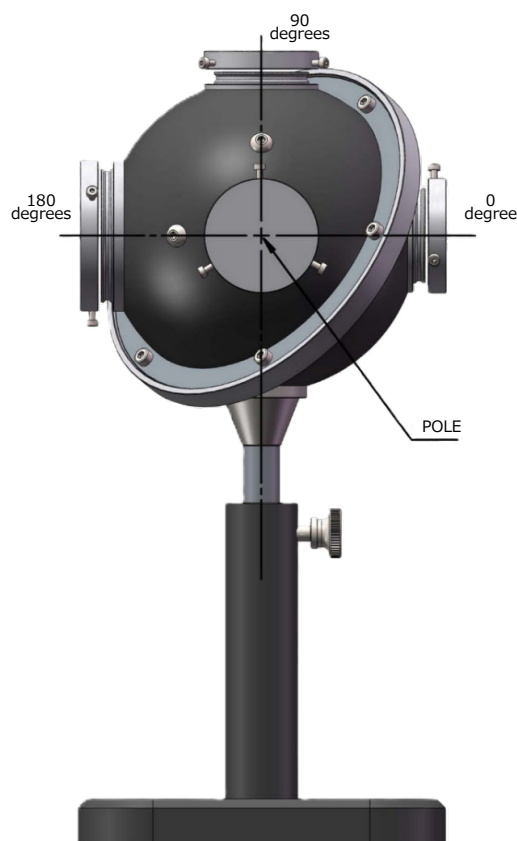
※The port and lid are standard equipment.



Features

1. White reflective surface with uniform and high reflectivity.
2. Even if the diffuse surface becomes contaminated or deteriorated, it can be restored and maintained.
3. Various light sources can be used by changing accessories such as sockets on the integrating sphere.
4. For general-purpose integrating spheres, we propose standard products with standard sizes and standard port numbers.

※ Our company can also accommodate custom orders and large integrating spheres up to 2 meters.



lid(COVER) Model number	Port Size (inches)	ΦA(mm)	ΦB(mm)	ΦC(mm)
NIS-CO-0.5	0.5	12.7	19	25
NIS-CO-1.0	1.0	25.4	32	38
NIS-CO-1.5	1.5	38.1	44	50
NIS-CO-2.0	2.0	50.8	57	63
NIS-CO-2.5	2.5	63.5	70	76

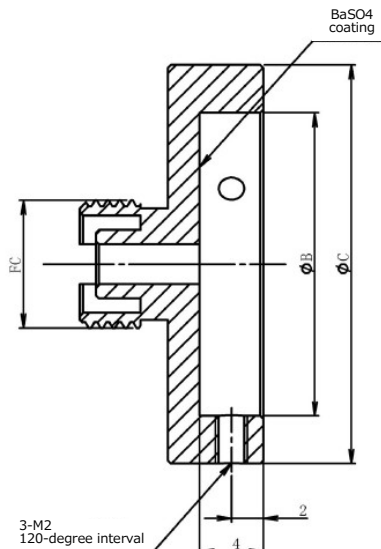
General-purpose integrating sphere

Additional options

FC Connector

FC Connector Model number	Port size (inches)	ΦA(mm)	ΦB(mm)	ΦC(mm)
NIS-FC-0.5	0.5	12.7	19	25
NIS-FC-1.0	1.0	25.4	32	38
NIS-FC-1.5	1.5	38.1	44	50
NIS-FC-2.0	2.0	50.8	57	63
NIS-FC-2.5	2.5	63.5	70	76

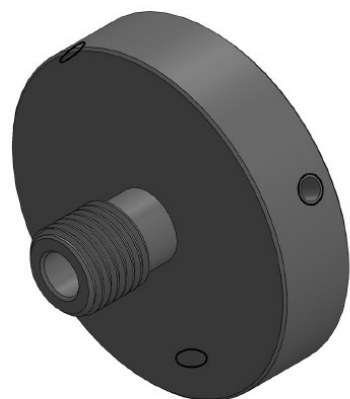
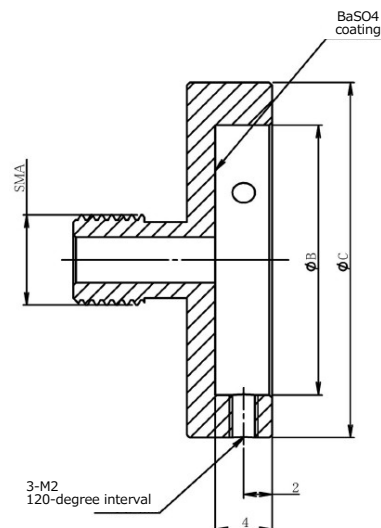
※Includes set screw



SMA Connector

SMA Connector Model number	Port size (inches)	ΦA(mm)	ΦB(mm)	ΦC(mm)
NIS-SMA-0.5	0.5	12.7	19	25
NIS-SMA-1.0	1.0	25.4	32	38
NIS-SMA-1.5	1.5	38.1	44	50
NIS-SMA-2.0	2.0	50.8	57	63
NIS-SMA-2.5	2.5	63.5	70	76

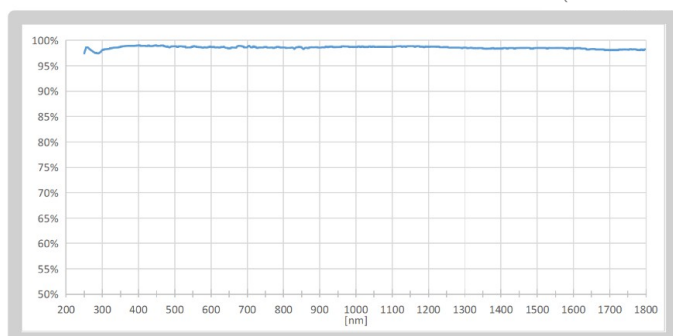
※Includes set screw



UV, visible, and infrared compatible - Diffuse reflection type White Reflectance Standard

This is a diffuse-reflectance standard white plate suitable for high-precision spectroscopic measurements. Made of sintered PTFE material, this product exhibits high reflectance and excellent diffusivity over a wide wavelength range of 250 nm to 2500 nm. It offers ideal performance for use as a reference reflective surface or for calibration purposes in integrating spheres, spectrometers, and optical evaluation systems.

(Nominal value)



Model number	NIS-PTFE-030	NIS-PTFE-050
Size	Φ30mm	Φ50mm
Wavelength	250~2500nm	250~2500nm
Heat resistance temperature	280°C	280°C
Case material	Aluminum	Aluminum